

Apoorv Mahajan

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SUMMARY

UX Researcher and UX Engineer with 6+ years of experience specializing in human-computer interaction (HCI) for AR/VR, AI, and automotive systems. Expert in conducting mixed-methods research to define interaction paradigms for multi-modal (voice/visual) interfaces. Proven track record of translating complex user behavioral data into high-fidelity prototypes and production-ready systems that prioritize safety, accessibility, and cognitive load management.

EDUCATION

2023 - 2025 Master of Science (Human-Computer Interaction) at **DePaul University** (GPA: 3.9/4.0)

2019 - 2023 Bachelor's Degree at **Symbiosis Center for Media and Communication** (GPA: 3.6/4.0)

WORK EXPERIENCE

UX Engineer | Woven by Toyota | Ann Arbor, MI

November 2025 - Present

- Lead UX Design for an AI Voice-first in-car experience for a compromised-driver demographic.
- Researched driver behavior, change in behavior after prototypes and iterated further. Created user flows, demo videos, mockups, working prototypes in Figma and Code, connected it to CARLA simulator for live UX Testing.
- Product is using Vision ML to Voice feedback with minimal visuals on the HMI (Ceter Screen) of the car providing live feedback and assistance to the driver.
- Engineered a custom Python backend bridge integrating the CARLA autonomous driving simulator with an iPad-based HMI web application (React, TypeScript, Bootstrap), achieving a 1-second visual glance-to-comprehension rate outperforming the 2-second automotive industry standard.
- Directed validation studies by triggering context-aware HMI screens and voice clips to test driver reactions to unexpected edge cases; proved that AI coaching increased driver confidence, improving overall driving efficiency by 33% and skill acquisition by 20% in initial sessions.
- Multi-modal feedback loops (voice-first and UI-based) compared to complex driving scenarios, optimizing speech parameters and interaction timing to successfully reduce measured driver cognitive load by 43%.
- Leveraged AI coding tools to rapidly script, install, and validate custom behavioral features and traffic environments in CARLA, significantly accelerating the deployment of high-fidelity human-interaction research scenarios.
- Designed a scalable design system that uses design tokens and visual regression testing to guaranty strict accessibility compliance, streamlined the Figma-to-CSS handoff process for the engineering team.
- Collaborating on a critical GAIA funded project with multiple Toyota companies (TRI, TMNA and TCNA).
- Daily Tools: Figma, VSCode, Claude Code, Python, PyGame, CARLA Simulator, GitHub, HTML, CSS, JS, DoveTail.

UI/UX Designer | Omnia | Chicago, IL

Sep 2024 - November 2025

- **Spatial Computing & XR Infrastructure:** Engineered high-performance 3D spatial interfaces for Apple Vision Pro and Meta Quest utilizing Unity (uGUI, TextMeshPro) and VisionOS SDKs, ensuring low-latency interaction and high visual fidelity across iOS, macOS, and XR environments.
- **Multimodal AI Validation:** Developed high-fidelity experimental prototypes in SwiftUI and React to A/B test AI-powered audible and visual feedback loops, successfully reducing user error by 13% during complex task execution in high-stakes automotive settings.
- **AR Usability & Product Strategy:** Directed formative usability testing for spatial interaction models and AR control schemes, refining visual hierarchies to reduce customer confusion by 13% and drive a 25% increase in subscriber retention.
- **Automotive UI/UX Engineering (ChiPark & Carvo):** Designed and deployed real-time geolocation applications for CarPlay and mobile ecosystems; integrated government-sourced APIs to reduce parking search times by 15% and increase payment speeds by 30%, while concurrently driving a 15% increase in fleet dispatch monitoring efficiency.

- **Safety-Critical Simulation:** Implemented Apple HIG, WCAG 2.0, and DMV requirements across all automotive UI components, utilizing Unity-based simulation prototypes to validate driver-assist motion, responsiveness, and distraction mitigation.
- **Spatial Training Prototyping:** Architected "How2To," an AI-powered VisionOS platform delivering real-time, step-by-step spatial tutorials, leading cross-functional engineering sprints to translate complex Figma prototypes into production-ready Unity assets.
- **VisionOS Fleet Management:** Prototyped advanced Vision Pro interfaces for spatial navigation, fleet management, and 3D object filtering to streamline swift asset tracking.

UX Researcher and UI Designer | Patched Reality | Chicago, IL

Aug 2024 - Sep 2024

- Reach Shoulder Health: Rebuilt Unity-driven VR healthcare interfaces (uGUI) for physiotherapy simulations, improving usability and interaction accuracy by 27 percent .
- Increased user retention among the elderly by 50 percent by introducing a level progression map
- Collaborated with developers to debug and profile Unity scene performance, ensuring smooth user interactions across hardware tiers
- Increased average physiotherapy session time by 20 percent by reducing UI related confusion
- Partnered with Unity engineers to align interface logic, user input handling, and rendering performance with UX goals
- SynCare: Conducted longitudinal user studies for AI-integrated EHR systems, identifying key friction points in medical data visualization that led to a 25 percent reduction in measured cognitive load.
- Concluded that motivational messaging to follow a prescribed diet increased healthy habits by 35 percent
- Reduced overall cognitive load caused by current EHR, medical, and pharmaceutical app flows by 25 percent

Founder and UX Consultant | Crazy Minds | Remote

Jul 2019 - Sep 2023

- Delivered 25+ UX/UI projects across fintech, health-tech, and e-commerce with measurable improvements
- Created modular design systems supporting developers' onboarding and faster product cycles
- Led user research and iterative redesign for mobile-first and web-first platforms
- Delivered Unity-based UI prototypes for AR/VR and SaaS clients, bridging front-end design with Unity interaction scripting for rapid iteration

LEADERSHIP AND COLLABORATION

- Planned, designed, collaborated and iterated on 3 B2B and 2 B2C products end-to-end with engineers to drive successful market launches
- Facilitated workshops and built trust with key stakeholders to align on product vision and implementation
- Developed and governed organization-wide Figma design systems (100+ reusable components), enforcing industry standards of consistency and reducing handoff/QA issues by over 25 percent
- Conducted development sprints using Jira and Confluence with developers for efficient usage of JS, HTML, CSS, React.js, Angular, Vue and SwiftUI to meet the design direction
- Presented design solutions to C-suite and senior leaders, communicating rationale and the impact of UI improvements
- Mentored visual design and UX Research interns

SKILLS

Research Methods: Quantitative & Qualitative Analysis, Longitudinal Studies, Contextual Inquiry, Eye-tracking Analysis

HCI Frameworks: Human Factors for Automotive, Spatial UI Heuristics, Cognitive Load Theory, WCAG 2.1/3.0.

Tools: Figma, SwiftUI, Storybook, Jira, Confluence

Development Collaboration: HTML, CSS, React.js, SwiftUI, UI Toolkit

Design Systems: WCAG, HIG, Modular Systems